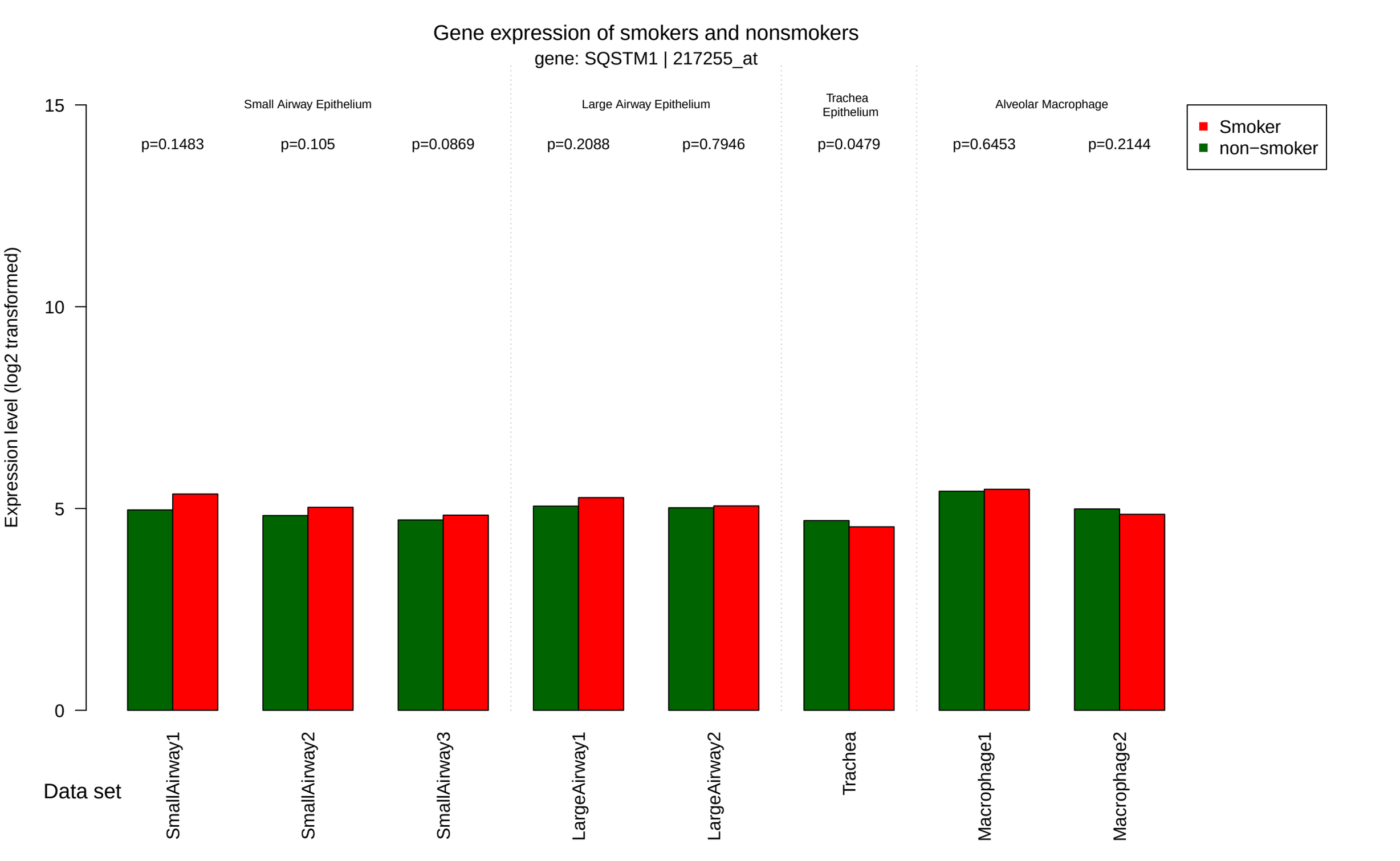
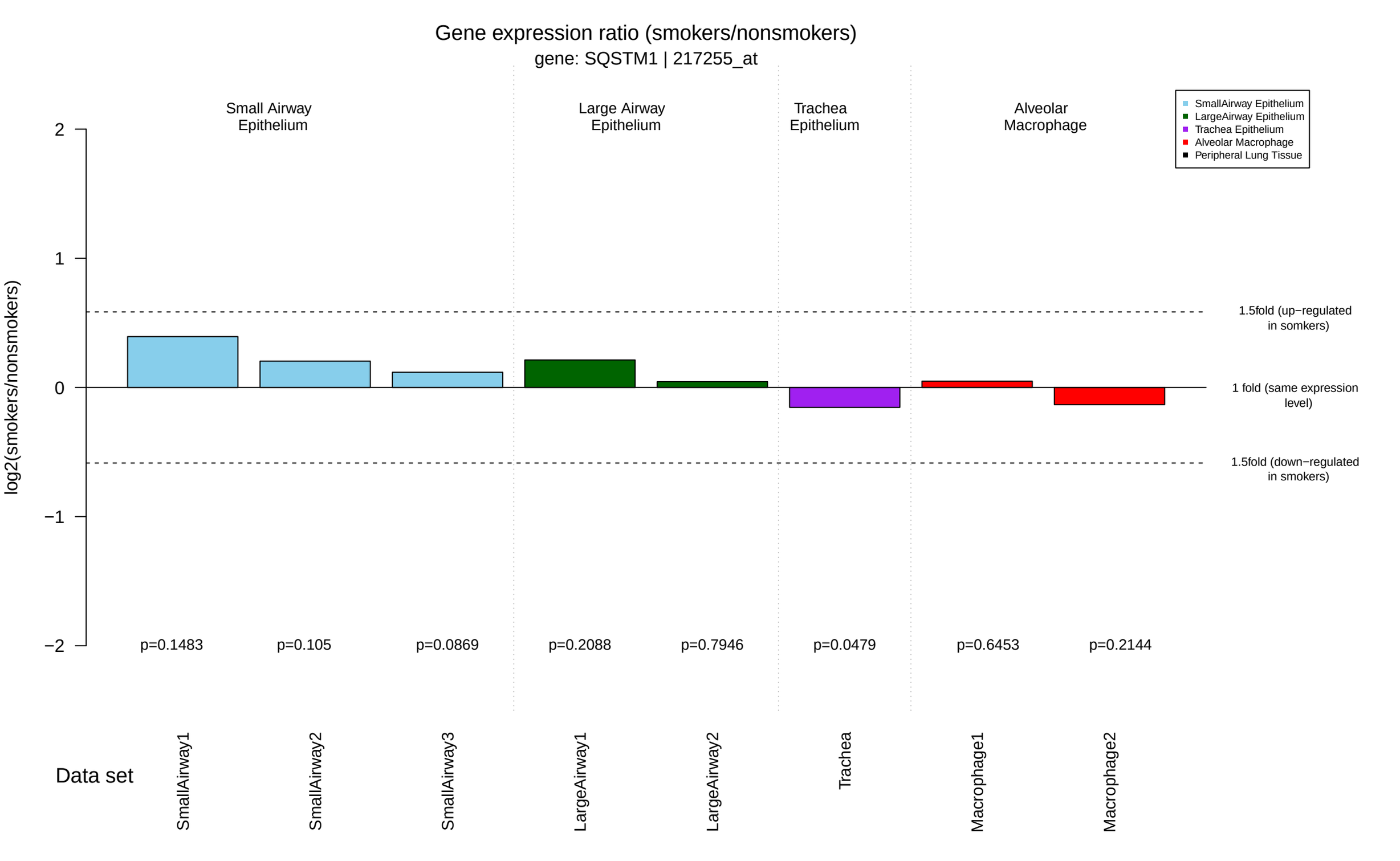
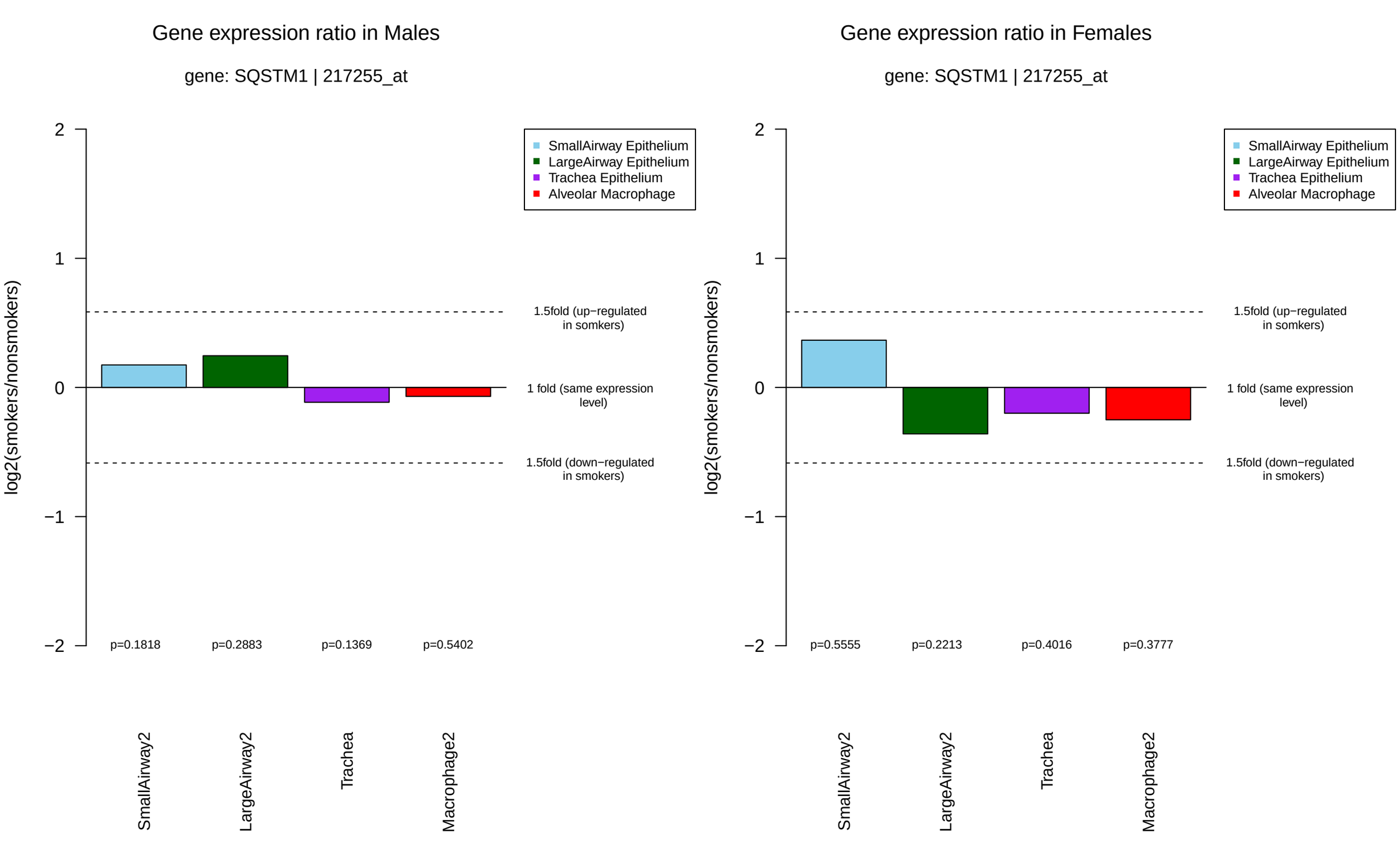
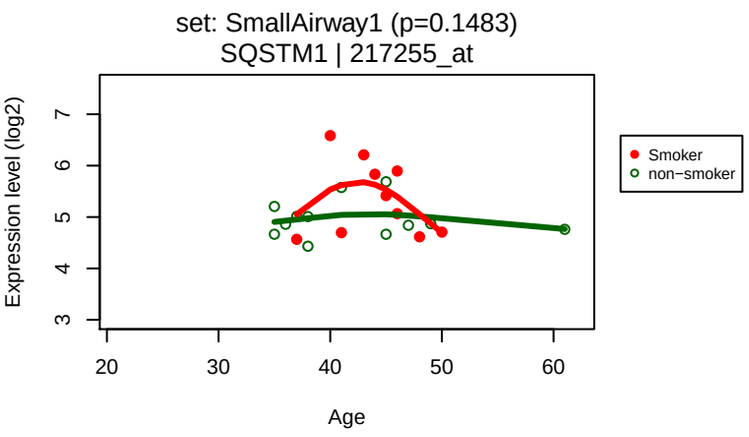
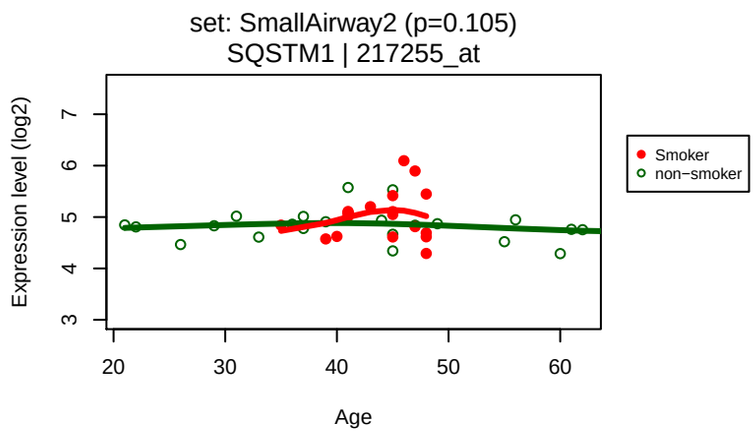
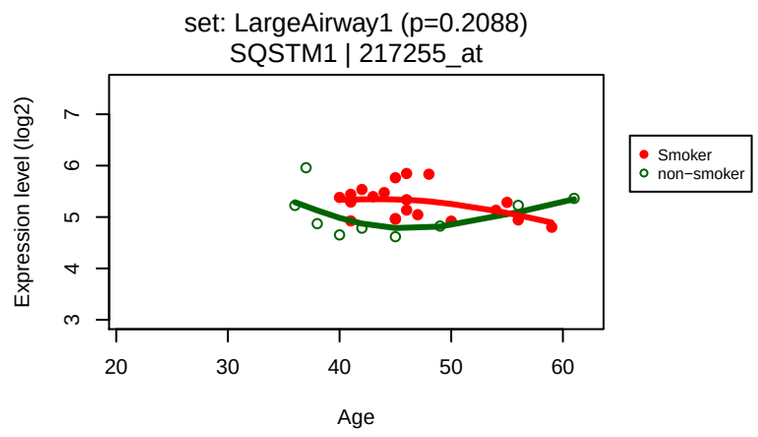

Gene expression of smokers and nonsmokers
gene: SQSTM1 | 217255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1483
p=0.105
p=0.0869
p=0.2088
p=0.7946
p=0.0479
p=0.6453
p=0.2144
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

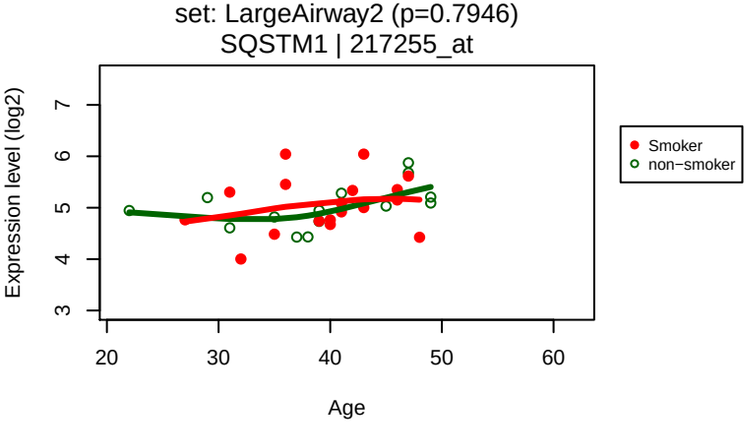

Gene expression ratio (smokers/nonsmokers)
gene: SQSTM1 | 217255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1483
p=0.105
p=0.0869
p=0.2088
p=0.7946
p=0.0479
p=0.6453
p=0.2144
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

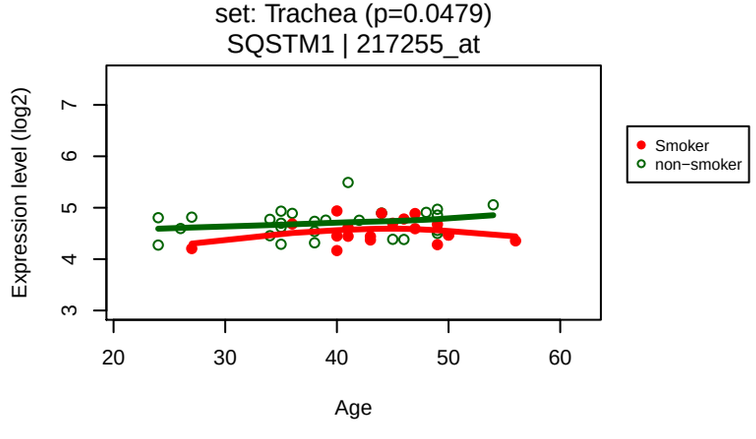

Gene expression ratio in Males
gene: SQSTM1 | 217255_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1818
p=0.2883
p=0.1369
p=0.5402
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SQSTM1 | 217255_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5555
p=0.2213
p=0.4016
p=0.3777
SmallAirway2
LargeAirway2
Trachea
Macrophage2

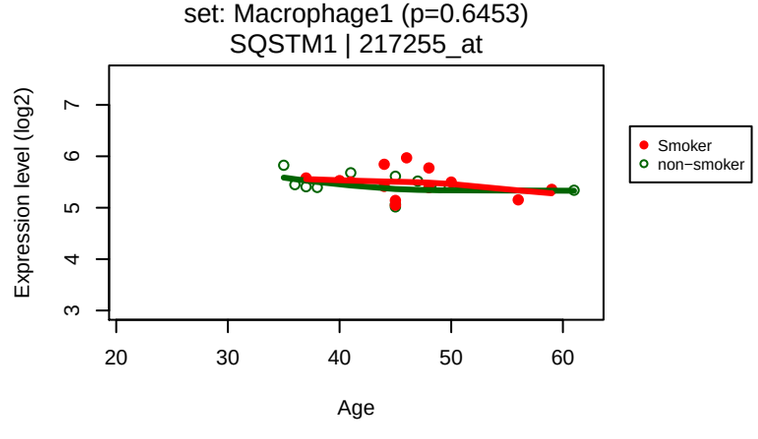

set: SmallAirway1 (p=0.1483)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker

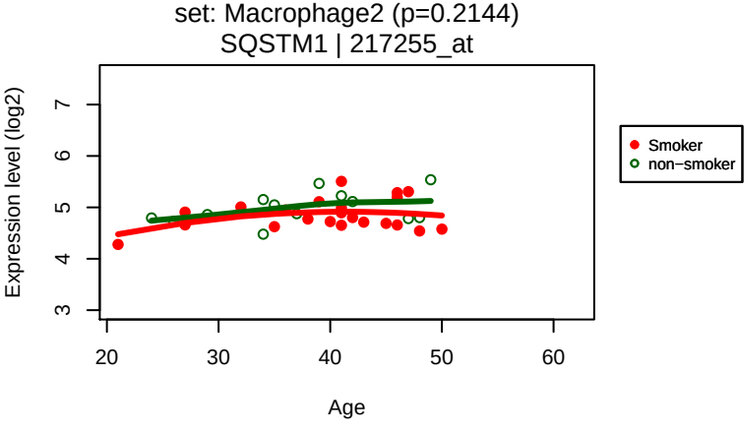

set: SmallAirway2 (p=0.105)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker

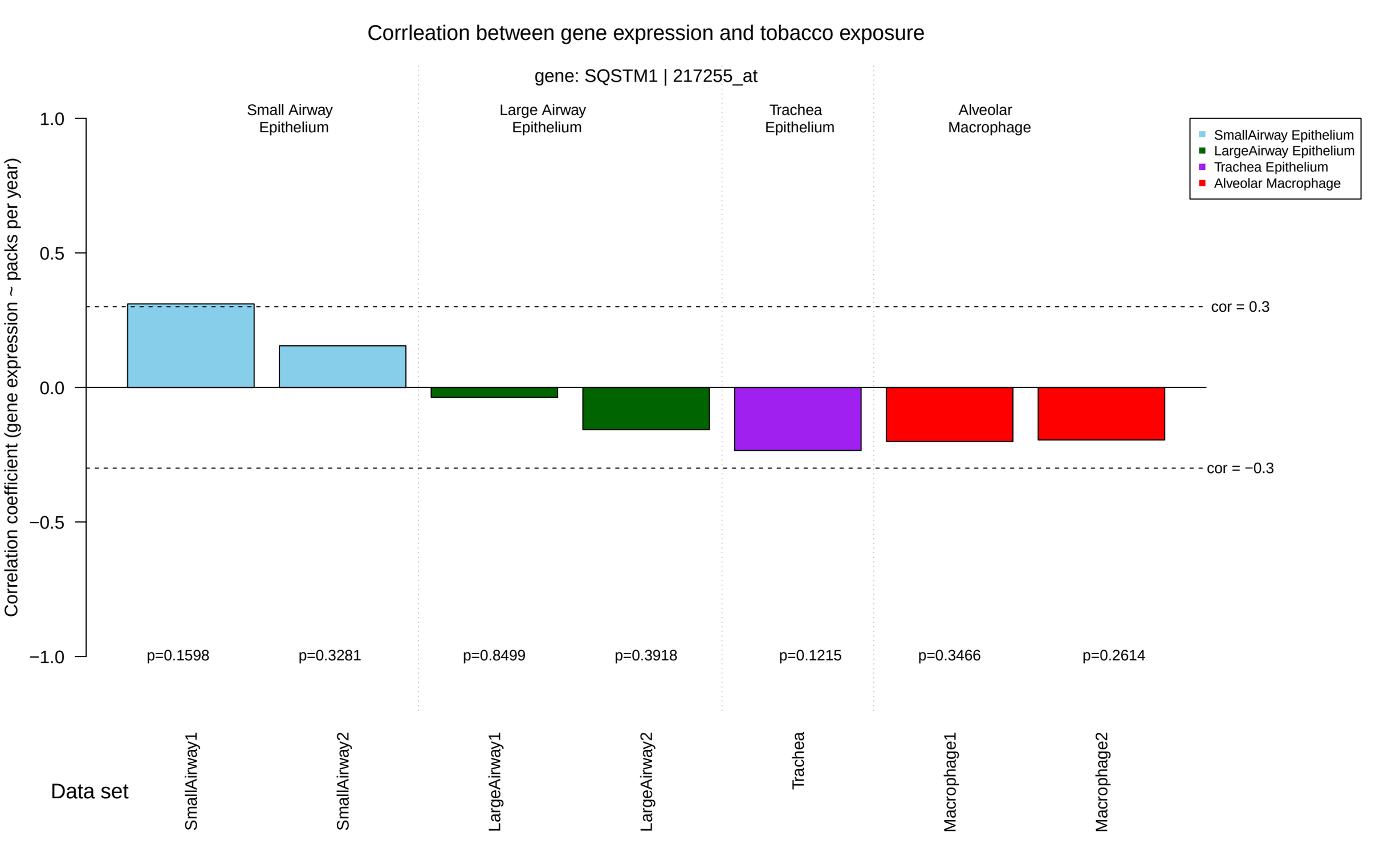

set: LargeAirway1 (p=0.2088)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7946)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0479)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6453)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2144)
SQSTM1 | 217255_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SQSTM1 | 217255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1598
p=0.3281
p=0.8499
p=0.3918
p=0.1215
p=0.3466
p=0.2614
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2